

Harvard College // Cambridge, MA // +1 (770)-733-4048 // rjun@college.harvard.edu // [LinkedIn](#)

PROFILE

Physics (Applied Physics Honors) junior focused on computational nanophotonics + polarimetric imaging. Experienced with RCWA/RETICOLO, Tidy3D/Lumerical, and Python/MATLAB for modeling and calibration.

EXPERIENCE

RESEARCH

Computational Neuroethology Unit // Okinawa Institute of Science & Technology // 2025 (Onsite)

- Collected polarimetric video dataset of cephalopods
- Processed DoLP/AoLP maps and annotated sequences to identify candidate signaling events

Capasso Group // Harvard University // 2021-2023 (HS, Hybrid), 2023-2024 (College, Onsite)

- Built RCWA/RETICOLO pipeline for metasurface response under partial coherence, enabled parameter sweeps with quadratic speedup compared to baseline
- Developed polarimetric calibration workflow (DoLP/AoLP) in Python using 24 targets, captured and processed polarimetric video sequences of cephalopods in collaboration with the Bellono Group

Publications

IEEE - R. Jun, et al., [Computing the Optical Response of Metasurfaces under Partially Coherent Illumination](#), Conference speaker at CLEO 2023

INDUSTRY

SK Signet EV America // Fairfax, VA, US // 2021

- Built comparison model of competitor charger specs (power levels, efficiency, thermal limits, uptime/MTBF claims, install requirements) to support product benchmarking

EDUCATION

HARVARD UNIVERSITY

A.B. Physics (Applied Physics Honors) // 2023-Expected 2027

Relevant Coursework: Optics and Photonics, Engineering Quantum Mechanics, Supervised Research, Systems and Control, Topics in Current Research (Physics)

GEORGIA INSTITUTE OF TECHNOLOGY

Dual Enrolled Student // 2021-2023

Relevant Coursework: Linear Algebra, Multivariable Calculus, Applied Combinatorics, Differential Equations

LEADERSHIP

PRESIDENT, TEAM MANAGER, STARTING PLAYER

Harvard eSports // 2024-Present

- Led independent student organization of ~300 registered members, co-managed budget of \$10,000+
- Brokered partnerships with major organizations (Red Bull, ExitLag, etc.) and managed deliverables

CO-PUBLICITY CHAIR + CHOREOGRAPHER

Harvard Asian American Dance Troupe // 2024-Present

- Co-led all social/publicity strategy (2024-2025) for troupe of 400+, produced content and releases
- Choreographed 5 pieces across semesterly sold-out shows with 600+ audience members, collaborated with performers and leadership

SKILLS

PROGRAMMING LANGUAGES

Python // Java (Oracle Certified Associate) // MATLAB // HTML // CSS // JavaScript // C#

SIMULATION/MODELING

RCWA // RETICOLO // FDTD // Tidy3d // Lumerical

AUDIO/VISUAL TOOLS

Premiere Pro // After Effects // Photoshop // Blender // FL Studio

LANGUAGES

Bilingual proficiency in Korean